

## Identification of an Unknown Copper (II) Salt

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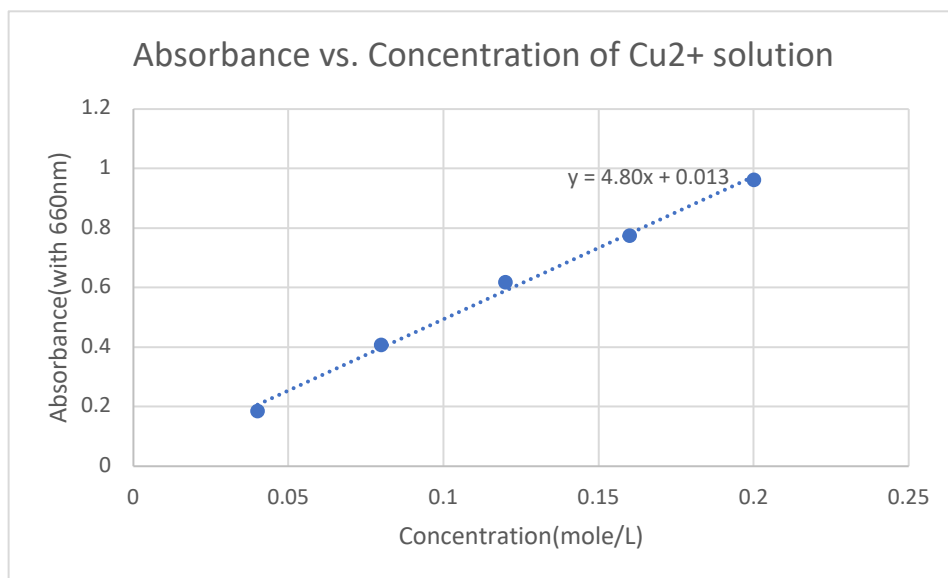
### 1. RESULT

The molar absorptivity (extinction coefficient) for  $\text{Cu}_2^+$  ions is  $\epsilon = 4.80 \text{ m}^2/\text{mol}$ . The absorbance of the solution of the unknown copper(II) salt with a wavelength of 660 nm is 0.785.

### 2. DISCUSSION

The unknown copper(II) salt is  $\text{Cu}(\text{BO}_2)_2$ . According to the obtained absorbance, absorptivity(extinction coefficient), and given  $l = 1 \text{ cm}$ , the molarity  $c$  of the solution can be determined by  $c = A/\epsilon l = 0.785/4.80 = 0.164 \text{ mol/L}$ . The mole of  $\text{Cu}_2^+$  in the solution can be calculated by Molarity  $\times$  Volume of solution ( $0.164 \text{ mol/L} \times 0.020 \text{ L} = 0.0033 \text{ mol}$ ). Given the gram of the unknown solid 0.50 g, the molecular weight can be determined by  $0.50 \text{ g}/0.0033 \text{ mol} = 150 \text{ g/mol}$ . Compared with the possible unknowns table given in the lab manual on page 30, the closest molecular weight of 150 g/mol is  $\text{Cu}(\text{BO}_2)_2$  (149 g/mol).

Therefore, the unknown copper(II) salt is  $\text{Cu}(\text{BO}_2)_2$ .



**Figure 1.** The independent variable of the equation is the molarity/concentration ( $\text{mole/L}$ ), the dependent variable is the Absorbance with the wavelength of 660 nm, the slope is the product of extinction coefficient  $\epsilon$  and path length ( $l = 1 \text{ cm}$  in this lab).